

MODELLING THE WINNER OF THE 2021/22 NBA SEASON

Possible **sample spaces**:

Person A

Los Angeles Lakers	0.33
Other	0.67

Person B

Los Angeles Lakers	0.3
Milwaukee Bucks	0.2
Other	0.5

Person C

Brooklyn Nets	0.25
Los Angeles Lakers	0.25
Milwaukee Bucks	0.15
Golden State Warriors	0.15
Other/no winner	0.2

Person D

Western Conference		Eastern Conference	
Los Angeles Lakers	0.2	Brooklyn Nets	0.25
Golden State Warriors	0.05	Milwaukee Bucks	0.1
Phoenix Suns	0.05	Philadelphia 76ers	0.03
Utah Jazz	0.05	Miami Heat	0.03
Los Angeles Clippers	0.03	Atlanta Hawks	0.02
Dallas Mavericks	0.03	Boston Celtics	0.02
Denver Nuggets	0.03	Chicago Bulls	0.01
Memphis Grizzlies	0.01	New York Knicks	0.01
Portland Trail Blazers	0.01	Indiana Pacers	0.01
San Antonio Spurs	0.01	Toronto Raptors	0.01
New Orleans Pelicans	0.01	Charlotte Hornets	0.01
Sacramento Kings	0.002	Washington Wizards	0.01
Oklahoma City Thunder	0.002	Orlando Magic	0.002
Houston Rockets	0.001	Cleveland Cavaliers	0.001
Minnesota Timberwolves	0.001	Detroit Pistons	0.001

QUESTIONS

For Person D, consider the following **events**:

$$E = \{\text{Teams in the Eastern Conference}\}$$

$$W = \{\text{Teams in the Western Conference}\}$$

$$A = \{\text{Teams whose name contains a reference to an animal}\}$$

$$T = \{\text{Team whose name (as above) contains three words}\}$$

$$C = \{\text{Teams based in California}\}$$

1. Determine the following:
 - a. $P(E)$
 - b. $P(W)$
 - c. $P(A)$
 - d. $P(T)$
 - e. $P(C)$
2. Determine also:
 - a. $P(W \cap A)$
 - b. $P(A \cap T)$
 - c. $P(A \cup T)$
3. Use the information in Questions 1 and 2 to determine $P(W \cup A)$. Check your answer by listing the elements of $W \cup A$ and summing their probabilities.
4. What is the relationship between events W and C ? Give your answer using set notation. Hence write down $P(W \cap C)$ and $P(W \cup C)$.
5. What is $A \cap C$? Give your answer using set notation. Write down $P(A \cap C)$ and determine $P(A \cup C)$.
6. Which other pairs of events given above are mutually exclusive?
7. Suppose that a time traveller tells Person D that the winner will be from the Western conference. Given this information (and assuming she believes it), with what probability should Person D model the chance that the winning team:
 - a. is based in California?
 - b. has a name that references an animal?
8. Suppose instead that the time traveller tells Person D that the winner has a name that references an animal. Given this information, with what probability should Person D model the chance the the winning team is from the Western conference?